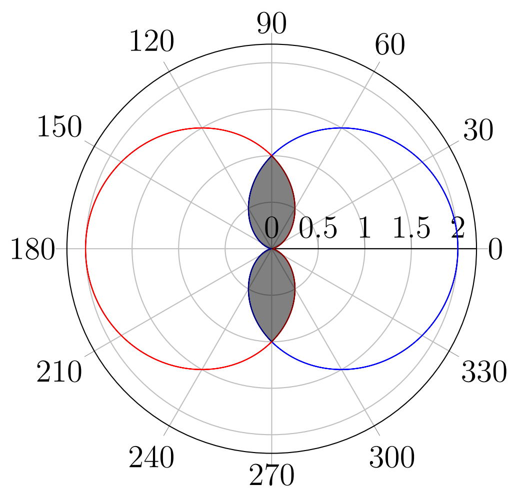
Math 2551 Worksheet: Polar Double Integrals

1. Evaluate $\iint_D y^2 + 3x \, dA$ where D is the region in the 3rd quadrant between $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$.

2. Change the Cartesian integral
$$\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-x^2-y^2} \, dx \, dy$$
into an equivalent polar integral and evaluate the integral.

3. Find the area of the region common to the interiors of the cardioids $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$.



4. Use a double integral to determine the volume of the solid that is inside the cylinder $x^2 + y^2 = 16$, below $z = 2x^2 + 2y^2$, and above the xy-plane.

5. **Challenge:** An integral of great importance in statistics is the Gaussian integral $I = \int_0^\infty e^{-x^2} \, dx$. The function $f(x) = e^{-x^2}$ has no elementary antiderivative, so this integral was hard to compute with the methods of single-variable calculus.

   Let $I^2 = \left(\int_0^\infty e^{-x^2} \, dx\right)\left(\int_0^\infty e^{-y^2} \, dy\right)$.

   (a) Express I^2 as the limit of a double integral in polar coordinates with an appropriately chosen domain. (Hint: As R goes to infinity, what happens to a disk of radius R centered at the origin?)

   (b) Evaluate your double interal to compute the value of I^2. Use this to find the value of the original Gaussian integral I.

You can find some history of this integral here.